

Coding & Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Coding & Solution

The **Emotion Detection & Learning Support Engine** is developed using **Python, Streamlit, Machine Learning, Natural Language Processing (NLP), and Generative AI** to detect users' emotions from text and provide personalized learning support. The system utilizes a fine-tuned **BERT emotion classification model** for accurate emotion prediction and integrates the **Google Gemini API** to deliver intelligent guidance and interactive responses. The project includes text preprocessing, model loading, emotion classification, personalized learning resource recommendations, and an intuitive web-based interface. The code follows a modular architecture with reusable components, proper error handling, and clean coding practices to ensure reliability, scalability, and maintainability.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/tanveerali06/Emotion-Detection-Learning-Support-Engine
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Streamlit, Transformers (Hugging Face), PyTorch, Pandas, NumPy, NLTK, Google Gemini API

Key Features Implemented	Emotion detection using BERT, AI-powered chatbot, personalized learning recommendations, text preprocessing, interactive Streamlit web interface, real-time emotion prediction
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Pending / Incomplete Features	User authentication, database integration, multilingual support, voice emotion recognition, cloud deployment, progress tracking dashboard
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , activate the virtual environment (if applicable), run <code>streamlit run app.py</code> , and open the local Streamlit URL (typically <code>http://localhost:8501</code>)

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository (GitHub)	Yes

Additional Notes / Comments:

The project follows a clean and modular coding structure with proper documentation, meaningful variable names, and error handling. It successfully detects user emotions, provides personalized learning recommendations, and offers AI-powered guidance through the Gemini API. The application is easy to maintain and can be enhanced with features such as user authentication, database integration, multilingual support, and cloud deployment in future versions.